

systems. Variations in mating system and mode of fertilization affect certainty of paternity, which in turn, through evolution, influences mating behavior and parental care.

- **Game theory** provides a way of thinking about evolution in situations where the fitness of a particular behavioral phenotype is influenced by other behavioral phenotypes in the population.

? In some spider species, the female eats the male immediately after copulation. How might you explain this behavior from an evolutionary perspective?

CONCEPT 51.4

Genetic analyses and the concept of inclusive fitness provide a basis for studying the evolution of behavior (pp. 1154–1160)

- Genetic studies in insects have revealed the existence of master regulatory genes that control complex behaviors. Within the underlying hierarchy, multiple genes influence specific behaviors, such as a courtship song. Research on voles illustrates how variation in a single gene can determine differences in complex behaviors.
- Behavioral variation within a species that corresponds to environmental variation may be evidence of past evolution.
- **Altruism** can be explained by the concept of **inclusive fitness**, the effect an individual has on proliferating its genes by producing its own offspring *and* by providing aid that enables close relatives to reproduce. The **coefficient of relatedness** and **Hamilton's rule** provide a way of measuring the strength of the selective forces favoring altruism against the potential cost of the “selfless” behavior. **Kin selection** favors altruistic behavior by enhancing the reproductive success of relatives.

? What insight about the genetic basis of behavior emerges from studying the effects of courtship mutations in fruit flies and of pair-bonding in voles?

TEST YOUR UNDERSTANDING

⇒ For more multiple-choice questions, go to the **Practice Test** in the **Mastering Biology** eText or Study Area, or go to goo.gl/GruWRg.

Levels 1-2: Remembering/Understanding

1. Which of the following is true of innate behaviors?
 - (A) Their expression is only weakly influenced by genes.
 - (B) They occur with or without environmental stimuli.
 - (C) They are expressed in most individuals in a population.
 - (D) They occur in invertebrates and some vertebrates but not mammals.
2. According to Hamilton's rule,
 - (A) natural selection does not favor altruistic behavior that causes the death of the altruist.
 - (B) natural selection favors altruistic acts when the resulting benefit to the recipient, corrected for relatedness, exceeds the cost to the altruist.
 - (C) natural selection is more likely to favor altruistic behavior that benefits an offspring than altruistic behavior that benefits a sibling.
 - (D) the effects of kin selection are larger than the effects of direct natural selection on individuals.

3. Female spotted sandpipers aggressively court males and, after mating, leave the clutch of young for the male to incubate. This sequence may be repeated several times with different males until no available males remain, forcing the female to incubate her last clutch. Which of the following terms best describes this behavior?
 - (A) polygyny
 - (B) polyandry
 - (C) promiscuity
 - (D) certainty of paternity

Levels 3-4: Applying/Analyzing

4. A region of the canary forebrain shrinks during the nonbreeding season and enlarges when breeding season begins. This change is probably associated with the annual
 - (A) addition of new syllables to a canary's song repertoire.
 - (B) crystallization of subsong into adult songs.
 - (C) sensitive period in which canary parents imprint on new offspring.
 - (D) elimination of the memorized template for songs sung the previous year.
5. Although many chimpanzees live in environments with oil palm nuts, members of only a few populations use stones to crack open the nuts. The likely explanation is that
 - (A) the behavioral difference is caused by genetic differences between populations.
 - (B) members of different populations have different nutritional requirements.
 - (C) the cultural tradition of using stones to crack nuts has arisen in only some populations.
 - (D) members of different populations differ in learning ability.
6. Which of the following is *not* required for a behavioral trait to evolve by natural selection?
 - (A) In each individual, the form of the behavior is determined entirely by genes.
 - (B) The behavior varies among individuals.
 - (C) An individual's reproductive success depends in part on how the behavior is performed.
 - (D) Some component of the behavior is genetically inherited.

Levels 5-6: Evaluating/Creating

7. **DRAW IT** You are considering two optimal foraging models for the behavior of a mussel-feeding shorebird, the oystercatcher. In model A, the energetic reward increases solely with mussel size. In model B, you take into consideration that larger mussels are more difficult to open. Draw a graph of reward (energy benefit on a scale of 0–10) versus mussel length (scale of 0–70 mm) for each model. Assume that mussels under 10 mm provide no benefit and are ignored by the birds. Also assume that mussels start becoming difficult to open when they reach 40 mm in length and impossible to open when 70 mm long. Considering the graphs you have drawn, indicate what observations and measurements you would want to make in this shorebird's habitat to help determine which model is more accurate.
8. **EVOLUTION CONNECTION** We often explain our behavior in terms of subjective feelings, motives, or reasons, but evolutionary explanations are based on reproductive fitness. Can both kinds of explanation be valid? For instance, is an explanation for behavior such as “falling in love” incompatible with an evolutionary explanation?